STAT 153 & 248 - Time Series
Lecture Twenty

Fall 2026, UC Berkeley

Aditya Guntuboyina

05 November 2026

In the last lecture, we started discussing Moving Average (MA) models. Given a positive integer $q \geq 1$, the Moving Average model with order q (denoted by MA(q)) is defined by the equation:

$$y_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_q \epsilon_{t-q} \tag{1}$$

where $\epsilon_t \overset{\text{i.i.d}}{\sim} N(0, \sigma^2)$. The MA($q$) model has $q+2$ unknown parameters which are estimated from observed data: $\mu, \theta_1, \ldots, \theta_q, \sigma$.

Here are some basic properties of the MA(q) model:

1. The mean of y_t is clearly equal to μ (so it is constant in t).
2. The covariance between y_t and y_{t+h} is given by:

$$\text{cov}(y_t, y_{t+h}) = \begin{cases} \sigma^2 \sum_{j=0}^{q-h} \theta_j \theta_{j+h} & : 0 \leq h \leq q \\ 0 & : h > q \end{cases}$$

The simplest of these MA(q) models is MA(1) (i.e., $q = 1$):

$$y_t = \mu + \epsilon_t + \theta \epsilon_{t-1}. \tag{2}$$

For the MA(1) model, we have

$$\text{correlation between } y_t \text{ and } y_{t+h} = \begin{cases} 1 & : h = 0 \\ \frac{\theta_1}{1+\theta_1^2} & : h = 1 \\ 0 & : h > 1 \end{cases}$$

In order to assess whether the MA(q) model is suitable for a particular dataset, a simple way is to compute the sample correlation between y_t and y_{t+h} for each value of h (in other words, the correlation between $(a_t, b_t), t = 1, \ldots, n-h$ where $a_t = y_t$ and $b_t = y_{t+h}$). This was the sample ACF plot that we discussed in the last lecture.

1 On Parameter Estimation in MA(q)

How to fit the parameters of MA(q)? We shall not study this in detail. We will simply rely on the `ARIMA` function for fitting these models to data. But here, I will illustrate the main

ideas in the simplest case $q = 1$ i.e., the MA(1) model. If we change the modeling assumption from $\epsilon_0, \ldots, \epsilon_n \overset{\text{i.i.d}}{\sim} N(0, \sigma^2)$ to $\epsilon_0 = 0, \epsilon_1, \ldots, \epsilon_n \overset{\text{i.i.d}}{\sim} N(0, \sigma^2)$, then it is quite easy to write the likelihood. This can also be interpreted as writing the conditional likelihood of the data $y_1, \ldots, y_n$ conditioned on $\epsilon_0 = 0$.

1.1 Conditional Likelihood for MA(1), conditioned on $\epsilon_0 = 0$

Instead of the full likelihood, the form of the conditional likelihood when conditioned on $\epsilon_0 = 0$ is simpler. This is similar to the story for the AR(1) model where the form of the likelihood when conditioned on y_1 is much simpler than the full likelihood. The difference here is that instead of conditioning on y_1 being the observed value, we are conditioning on $\epsilon_0 = 0$.

Below we write the conditional likelihood:

$$f_{y_1,\ldots,y_n|\epsilon_0=0}(y_1, \ldots, y_n).$$

This likelihood can be broken down as

$$f_{y_1|\epsilon_0=0}(y_1) f_{y_2|y_1,\epsilon_0=0}(y_2) f_{y_3|y_1,y_2,\epsilon_0=0}(y_3) \cdots f_{y_n|y_1,\ldots,y_{n-1},\epsilon_0=0}(y_n)$$

Each of the terms above can be written explicitly. Let $\hat{\epsilon}_1 = y_1 - \mu$ and, for $t = 2, \ldots, n$,

$$\hat{\epsilon}_t = y_t - \mu - \theta \hat{\epsilon}_{t-1}$$

Then

$$f_{y_1|\epsilon_0=0}(y_1) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_1 - \mu)^2}{2\sigma^2}\right)$$

and

$$\begin{aligned} f_{y_t|y_1,\ldots,y_{t-1},\epsilon=0}(y_t) &= f_{y_t|\epsilon_1=\hat{\epsilon}_1,\epsilon_2=\hat{\epsilon}_2,\ldots,\epsilon_{t-1}=\hat{\epsilon}_{t-1},\epsilon_0=0}(y_t) \\ &= f_{\epsilon_t}(y_t - \mu - \theta\hat{\epsilon}_{t-1}) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(y_t - \mu - \theta\hat{\epsilon}_{t-1})^2\right). \end{aligned}$$

Thus the conditional likelihood given $\epsilon_0 = 0$ is given by

$$\begin{aligned} \text{likelihood} &= \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_1 - \mu)^2}{2\sigma^2}\right) \left[\prod_{t=2}^{n} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(y_t - \mu - \theta\hat{\epsilon}_{t-1})^2\right)\right] \\ &= \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\hat{\epsilon}_1^2}{2\sigma^2}\right) \left[\prod_{t=2}^{n} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\hat{\epsilon}_t^2}{2\sigma^2}\right)\right] \\ &= \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^n \exp\left(-\frac{1}{2\sigma^2}\sum_{t=1}^{n} \hat{\epsilon}_t^2\right). \end{aligned}$$

The key term in the above likelihood is $\sum_{t=1}^{n} \hat{\epsilon}_t^2$ which depends on the data $\{y_t\}$ and also on the parameters μ and θ. So let us denote

$$S(\mu, \theta) := \sum_{t=1}^{n} \hat{\epsilon}_t^2$$

so that the likelihood and log-likelihood become

$$\text{likelihood} = \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^n \exp\left(-\frac{S(\mu, \theta)}{2\sigma^2}\right)$$

and

$$\text{log-likelihood} = -\frac{n}{2}\log(\sigma^2) - \frac{S(\mu,\theta)}{2\sigma^2}.$$

Thus the MLEs of μ, θ are obtained by minimizing $S(\mu,\theta)$:

$$(\hat{\mu},\hat{\theta}) = \underset{\mu,\theta}{\text{argmin}}\ S(\mu,\theta).$$

We can use numerical optimization techniques to obtain $\hat{\mu},\hat{\theta}$. For example, in python, scipy has functions for minimization that can be directly used (with some reasonable initialization such at $\mu_{\text{init}} = \bar{y}$ and $\theta_{\text{init}} = 0$).

After obtaining $\hat{\mu}$ and $\hat{\theta}$, the MLE of σ is obtained as

$$\hat{\sigma} = \sqrt{\frac{S(\hat{\mu},\hat{\theta})}{n}}$$

After obtaining the MLEs $\hat{\mu}$ and $\hat{\theta}$, the next step is to obtain standard errors. For this, we can use the Bayesian approach with the standard prior:

$$\mu, \theta, \log\sigma \overset{\text{i.i.d}}{\sim} \text{unif}(-C, C).$$

The posterior is then

$$\begin{aligned} f_{\mu,\theta,\sigma|\text{data}}(\mu,\theta,\sigma) &\propto \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^n \exp\left(-\frac{S(\mu,\theta)}{2\sigma^2}\right) \times \frac{1}{\sigma} I\{-C < \mu,\theta,\log\sigma < C\} \\ &\propto \sigma^{-n-1} \exp\left(-\frac{S(\mu,\theta)}{2\sigma^2}\right) I\{-C < \mu,\theta,\log\sigma < C\} \end{aligned}$$

This is the joint posterior of μ, θ, σ. To obtain the posterior of μ, θ (without σ), we integrate the above over σ (from 0 to ∞). This integration is exactly the same as Lecture 4, and we get

$$f_{\mu,\theta|\text{data}}(\mu,\theta) \propto \left(\frac{1}{S(\mu,\theta)}\right)^{n/2} I\{-C < \mu,\theta < C\} \propto \left(\frac{S(\hat{\mu},\hat{\theta})}{S(\mu,\theta)}\right)^{n/2} I\{-C < \mu,\theta < C\}$$

If $S(\mu,\theta)$ is a quadratic function of μ, θ, then this will be a t-density. However our $S(\mu,\theta)$ will involve higher powers of θ and is not quadratic. The posterior usually is fairly concentrated around the MLE $(\hat{\mu},\hat{\theta})$ though, so we it makes sense to approximate $S(\mu,\theta)$ by a quadratic around $(\hat{\mu},\hat{\theta})$. This is done by Taylor expansion as follows. Let $\alpha = (\mu,\theta)$ and $\hat{\alpha} = (\hat{\mu},\hat{\theta})$. Taylor expansion for α near $\hat{\alpha}$ gives

$$\begin{aligned} S(\alpha) &= S(\hat{\alpha}) + \langle \nabla S(\hat{\alpha}), \alpha - \hat{\alpha}\rangle + (\alpha-\hat{\alpha})^T \left(\frac{1}{2} HS(\hat{\alpha})\right)(\alpha - \hat{\alpha}) \\ &= S(\hat{\alpha}) + (\alpha-\hat{\alpha})^T \left(\frac{1}{2} HS(\hat{\alpha})\right)(\alpha - \hat{\alpha}) \end{aligned}$$

where we used $\nabla S(\hat{\alpha}) = 0$ because $\hat{\alpha}$ minimizes $S(\alpha)$. Here $HS(\hat{\alpha})$ denotes the Hessian of S at $\hat{\alpha}$.

Therefore

$$
\begin{aligned}
& f_{\mu,\theta|\text{data}}(\mu,\theta) \\
& \propto \left(\frac{1}{S(\mu,\theta)}\right)^{n/2} I\{-C<\mu,\theta<C\} \\
& \propto \left(\frac{S(\hat{\alpha})}{S(\alpha)}\right)^{n/2} I\{-C<\mu,\theta<C\} \\
& = \left(\frac{S(\hat{\alpha})}{S(\hat{\alpha})+(\alpha-\hat{\alpha})^T\left(\frac{1}{2}HS(\hat{\alpha})\right)(\alpha-\hat{\alpha})}\right)^{n/2} I\{-C<\mu,\theta<C\} \\
& = \left(\frac{1}{1+(\alpha-\hat{\alpha})^T\left(\frac{1}{2S(\hat{\alpha})}HS(\hat{\alpha})\right)(\alpha-\hat{\alpha})}\right)^{n/2} I\{-C<\mu,\theta<C\} \\
& = \left(\frac{1}{1+\frac{1}{n-2}(\alpha-\hat{\alpha})^T\left(\frac{n-2}{2S(\hat{\alpha})}HS(\hat{\alpha})\right)(\alpha-\hat{\alpha})}\right)^{\frac{n-2+2}{2}} I\{-C<\mu,\theta<C\}.
\end{aligned}
$$

Comparing the above with the formula:

$$\left(\frac{1}{1+\frac{1}{k}(x-m)^T\Sigma^{-1}(x-m)}\right)^{\frac{k+p}{2}}$$

for the p-variate t-density $t_{k,p}(\mu,\Sigma)$, we see that (ignoring the indicator function $I\{-C<\mu,\theta<C\}$)

$$\alpha \mid \text{data} \sim t_{n-2,2}\left(\hat{\alpha}, \frac{S(\hat{\alpha})}{n-2}\left(\frac{1}{2}HS(\hat{\alpha})\right)^{-1}\right).$$

So the standard errors corresponding to μ and θ can be obtained by taking the square roots of the diagonal entries of

$$\frac{S(\hat{\alpha})}{n-2}\left(\frac{1}{2}HS(\hat{\alpha})\right)^{-1}$$

In order to compute these, we need to calculate the Hessian of $S(\alpha)$ with respect to α. We will use numerical differentiation (specifically a function from the library `numdifftools`) for this.

1.2 Full Likelihood for MA(1)

As we have seen in the previous section, the likelihood is easy to write down if we make the assumption $\epsilon_0 = 0$. Here we discuss how to write the original likelihood.

The joint density of $y_1,\ldots,y_n$ is multivariate normal with mean vector $m := (\mu,\ldots,\mu)^T$ and covariance matrix Σ where Σ equals the $n\times n$ matrix whose $(i,j)^{th}$ entry is given by

$$\Sigma(i,j) = \begin{cases} \sigma^2\left(1+\theta^2\right) & \text{when } i=j \\ \sigma^2\theta & \text{when } |i-j|=1 \\ 0 & \text{for all other } (i,j) \end{cases}$$

The likelihood is therefore

$$\left(\frac{1}{\sqrt{2\pi}}\right)^n (\det \Sigma)^{-1/2} \exp\left(-\frac{1}{2}(y-m)'\Sigma^{-1}(y-m)\right)$$

where y is the $n \times 1$ vector with components $y_1, \ldots, y_n$. This is a function of the unknown parameters μ, θ, σ which can be estimated by maximizing the logarithm of the likelihood. The presence of Σ^{-1} makes this computationally expensive.

A more explicit formula for the full likelihood can be written using the algorithm described below. This is a special case of the Kalman filter (see e.g., `https://en.wikipedia.org/wiki/Kalman_filter`). We will not be studying the Kalman filter in any generality in this course (a standard reference if you are interested is Durbin and Koopman [1]).

The basic first step is to write

$$\begin{aligned} f_{y_1,\ldots,y_n}(y_1,\ldots,y_n) &= f_{y_1}(y_1) f_{y_2|y_1}(y_2) f_{y_3|y_1,y_2}(y_3) \ldots f_{y_n|y_1,\ldots,y_{n-1}}(y_n) \\ &= \prod_{t=1}^{n} f_{y_t|y_1,\ldots,y_{t-1}}(y_t) \end{aligned}$$

where the conditional density $f_{y_t|y_1,\ldots,y_{t-1}}(y_t)$ should be interpreted as $f_{y_1}(y_1)$ when $t = 1$.

Each conditional density $y_t \mid y_1, \ldots, y_{t-1}$ is normal (because $y_1, \ldots, y_n$ is multivariate normal), all we need to do is to figure out the mean and variance of the density of $y_t \mid y_1, \ldots, y_{t-1}$. Let

$$a_t = \mathbb{E}(y_t \mid y_1, \ldots, y_{t-1}) \quad \text{and} \quad V_t = \operatorname{var}(y_t \mid y_1, \ldots, y_{t-1}).$$

We will figure out (a_t, V_t) sequentially for $t = 1, \ldots, n$. When $t = 1$, we have

$$a_1 = \mathbb{E}(y_1) = \mathbb{E}(\mu + \epsilon_1 + \theta\epsilon_0) = \mu \quad \text{and} \quad V_1 = \operatorname{var}(y_1) = \operatorname{var}(\mu + \epsilon_1 + \theta\epsilon_0) = \sigma^2(1+\theta^2).$$

We now derive a formula for computing a_t, V_t given a_{t-1}, V_{t-1} (for each $t \geq 2$). For this, we use the following fact:

$$\begin{pmatrix} y_t \\ y_{t-1} \end{pmatrix} \mid y_1, \ldots, y_{t-2} \sim N\left(\begin{pmatrix} \mu \\ a_{t-1} \end{pmatrix}, \begin{pmatrix} \sigma^2(1+\theta^2) & \theta\sigma^2 \\ \theta\sigma^2 & V_{t-1} \end{pmatrix}\right) \tag{3}$$

The proof of (3) is given below. Using (3), it is easy to see (using the formula for conditional distributions corresponding to a bivariate normal distribution) that

$$y_t \mid y_1, \ldots, y_{t-2}, y_{t-1} \sim N\left(\mu + \frac{\theta\sigma^2}{V_{t-1}}(y_{t-1} - a_{t-1}), \sigma^2(1+\theta^2) - \frac{\theta^2\sigma^4}{V_{t-1}}\right).$$

Thus

$$a_t = \mu + \frac{\theta\sigma^2}{V_{t-1}}(y_{t-1} - a_{t-1}) \quad \text{and} \quad V_t = \sigma^2(1+\theta^2) - \frac{\theta^2\sigma^4}{V_{t-1}}. \tag{4}$$

Therefore the algorithm for computing $(a_t, V_t), t = 1, \ldots, n$ is:

1. Set $a_1 = \mu$ and $V_1 = \sigma^2(1+\theta^2)$.
2. For $t = 2, \ldots, n$, given a_{t-1}, V_{t-1}, compute a_t, V_t via (4).

After calculating $(a_t, V_t), t = 1, \ldots, n$, the likelihood becomes

$$\text{likelihood} = \prod_{t=1}^{n} \frac{1}{\sqrt{2\pi V_t}} \exp\left(-\frac{(y_t - a_t)^2}{2V_t}\right).$$

The `ARIMA` function in `statsmodels` computes the MLE based on this likelihood.

Here is the proof of (3). We need to calculate the required conditional means, variances and covariances. Just observe

$$\mathbb{E}(y_t \mid y_1, \ldots, y_{t-2}) = \mathbb{E}(\mu + \epsilon_t + \theta\epsilon_{t-1} \mid y_1, \ldots, y_{t-2}) = \mu$$

because ϵ_t and ϵ_{t-1} are independent of $y_1, \ldots, y_{t-2}$, and

$$\mathbb{E}(y_{t-1} \mid y_1, \ldots, y_{t-2}) = a_{t-1}.$$

Further

$$\mathrm{var}(y_t \mid y_1, \ldots, y_{t-2}) = \mathrm{var}(\mu + \epsilon_t + \theta\epsilon_{t-1} \mid y_1, \ldots, y_{t-2}) = \mathrm{var}(\epsilon_t + \theta\epsilon_{t-1}) = \sigma^2(1+\theta^2),$$

and

$$\begin{aligned}
&\mathrm{cov}(y_t, y_{t-1} \mid y_1, \ldots, y_{t-2}) \\
&= \mathrm{cov}\left(\epsilon_t + \theta\epsilon_{t-1}, \epsilon_{t-1} + \theta\epsilon_{t-2} \mid y_1, \ldots, y_{t-2}\right) \\
&= \mathrm{cov}(\epsilon_t, \epsilon_{t-1} \mid y_1, \ldots, y_{t-2}) + \theta\mathrm{var}(\epsilon_{t-1} \mid y_1, \ldots, y_{t-2}) \\
&+ \theta\mathrm{cov}(\epsilon_t, \epsilon_{t-2} \mid y_1, \ldots, y_{t-2}) + \theta^2\mathrm{cov}(\epsilon_{t-1}, \epsilon_{t-2} \mid y_1, \ldots, y_{t-2}) \\
&= 0 + \theta\sigma^2 + 0 + 0 = \theta\sigma^2.
\end{aligned}$$

To see why the last covariance equals zero, just note

$$\mathrm{cov}(\epsilon_{t-1}, \epsilon_{t-2} \mid y_1, \ldots, y_{t-2}) = \mathbb{E}\left(\epsilon_{t-1}\epsilon_{t-2} \mid y_1, \ldots, y_{t-2}\right).$$

The above equals zero because

$$\mathbb{E}\left(\epsilon_{t-1} \mid \epsilon_{t-2}, y_1, \ldots, y_{t-2}\right) = 0.$$

Finally

$$\mathrm{var}(y_{t-1} \mid y_1, \ldots, y_{t-2}) = V_{t-1}.$$

This completes the proof of (3).

References

[1] Durbin, J. and S. J. Koopman (2012). *Time series analysis by state space methods* (Second ed.). Oxford university press.